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At a particular instant, the concentration of H_2, $I_2$and HI are found to be 0.2 mol L^{-1}, 0.2 mol L^{-1} and 0.6 mol L^{-1} respectively. From the above information we can predict the direction of reaction as follows.

$$Q = \frac{[HI]^2}{[H_2][I_2]} = \frac{0.6 \times 0.6}{0.2 \times 0.2} = 9$$

Since $Q < K_c$, the reaction will proceed in the forward direction.

Example 2

The value of Kc for the reaction

$$N_2O_4\,(g) \rightleftharpoons 2NO_2(g)$$

Kc = 0.21 at 373 K. The concentrations N_2O_4 and NO_2 are found to be 0.125 mol dm^{-3} and 0.5 mol dm^{-3} respectively at a given time. From the above information we can predict the direction of reaction as follows.

$$Q = \frac{[NO_2]^2}{[N_2O_4]} = \frac{0.5 \times 0.5}{0.125} = 2$$

The Q value is greater than Kc. Hence, the reaction will proceed in the reverse direction until the Q value reaches 0.21

Evaluate Yourself

3) The following water gas shift reaction is an important industrial process for the production of hydrogen gas.

$$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$$

At a given temperature K_p = 2.7. If 0.13 mol of CO, 0.56 mol of water, 0.78 mol of CO_2 and and 0.28 mol of H_2 are introduced into a 2 L flask, and find out in which direction must the reaction proceed to reach equilibrium

8.7.3 Calculation of concentration of reactants and products at equilibrium

If the equilibrium concentrations of reactants and products are known for a reaction, then the equilibrium constant can be calculated and vice versa.

Let us consider the formation of HI in which, 'a' moles of hydrogen and 'b' moles of iodine gas are allowed to react in a container of volume V. Let 'x' moles of each of H_2 and I_2 react together to form 2x moles of HI.

$$H_2\,(g) + I_{2\,(g)} \rightleftharpoons 2HI_2(g)$$

	H_2	I_2	HI
Initial number of moles	a	b	0
number of moles reacted	x	x	0
Number of moles at equilibrium	a-x	b-x	2x
Active mass or molar concentration at equilibrium	$\frac{a-x}{V}$	$\frac{b-x}{V}$	$\frac{2x}{V}$

Applying law of mass action,

$$K_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{\left(\frac{2x}{V}\right)^2}{\left(\frac{a-x}{V}\right)\left(\frac{b-x}{V}\right)} = \frac{4x^2}{(a-x)(b-x)}$$

The equilibrium constant K_p can also be calculated as follows:

We know the relationship between the K_c and K_p

$$K_P = K_C\,(RT)^{(\Delta n_g)}$$